

Public Cloud PostgreSQL Benchmarking Roadmap

Status: WIP. Last updated: June 16, 2026.

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Background

Spare Cores already provides cloud server specs, pricing, performance, and cost-efficiency data to SoftwareOne, powering their pre-sales tool (Cloud Performance Lab).

SoftwareOne, in collaboration with Microsoft and AMD, wishes to extend the Cloud Performance Lab to managed databases (DBaaS) as well, and subcontracts Spare Cores to design and implement externally credible data to support the following sales discussions:

- Migrating an on-prem database to Azure DBaaS.
- Migrating a self-hosted database already in Azure to Azure DBaaS.
- Migrating a self-hosted database from another cloud vendor to Azure DBaaS.
- Migrating a DBaaS from another cloud vendor to Azure DBaaS.

The Spare Cores team has the technical knowledge and capacity to

- Extend the [Navigator](#) (database and API) schemas to support DBaaS products besides the existing cloud server-focused interfaces.
- Spec out and implement the orchestration layer of mass execution of server<>client benchmarks across both DBaaS and IaaS products of cloud providers.
- Integrate the database performance benchmarks in the Navigator.
- Integrate the DBaaS qualitative features into the Navigator.
- Support SoftwareOne with the updated Navigator integration.

But seeks external support (likely from benchANT) to

- Design and validate the database benchmarking methodology – including database configurations and their auto-tuning for various instance sizes, benchmark client/data/query selection etc.
- Refresh and extend the qualitative data collected on DBaaS products as a continuation of their [DBaaS Navigator](#).

Project deliveries:

1. Performance benchmark scores in Spare Cores Navigator.
2. Qualitative DBaaS features in Spare Cores Navigator.
3. SoftwareOne Cloud Performance Lab interface for DBaaS sales conversations.
4. Whitepaper vetted by all parties (including Microsoft and AMD).

Project delivery deadline: Aug 31, 2026.

Responsibilities

SoftwareOne

Team: Frederik, Simon, Ali, others?

Tasks:

- Overall project management.
- Negotiate cloud credits or budget for benchmarking.
- Integrating the Spare Cores Navigator updates.
- Vet results.
- Cloud Performance Lab interface.
- Writing the whitepaper.

Microsoft

Tasks:

- Technical support.
- Vet results and whitepaper.
- Cloud credits? Spare Cores has around \$20k credits expiring in July because larger machines cannot be started using promotional credits as per our quotas and AM. Ideally, Microsoft resolves our quota issues and extends the validity of the cloud credits, otherwise we will have to pay for the instances and databases out of pocket.

AMD

Tasks:

- Technical support.
- Vet results and whitepaper.

Spare Cores

Status: to be finalized, but will be a subcontractor under SoftwareOne.

Team: Gergely (project management, Q&A), Attila (infra), Tamas (Navigator DB schemas and API), Alex (data analysis)

Tasks:

- Navigator database schema updates:
 - New “database” table for DBaaS products including features such as database engine, TLS/SSL support, disk encryption, basic/advanced monitoring, query logs, SLA and response times etc.
 - New “database_price” table for regional prices of the DBaaS products.
 - Updated “benchmark_scores” to support “database” type besides “server”.
- Navigator API updates as per the database schema changes.

- Ingest qualitative DBaaS features into the Navigator database.
- Pulumi templates for all supported vendors to start multiple managed databases and/or instances (such as “1 database server” and “1 benchmarking client”) in groups, as close to each other as possible, and over a private network to reduce latency (e.g. “cluster” placement group in AWS).
 - Vendor coverage:
 - Required:
 - Azure (both self-hosted and managed databases).
 - Further vendors (min 2), sorted by importance (TBD by SoftwareOne) – depending on budget and timeline:
 - AWS (both self-hosted and managed databases).
 - GCP (both self-hosted and managed databases).
 - OVH (both self-hosted and managed databases).
 - Vultr (both self-hosted and managed databases).
 - Alibaba Cloud (both self-hosted and managed databases).
 - UpCloud (both self-hosted and managed databases).
 - Hetzner Cloud (self-hosted).
- Docker image for a self-hosted PostgreSQL database, including auto-tuning of its configuration file for the host resources (e.g. based on available vCPUs and memory – TBD after collab with benchANT), and database records for benchmarking. This will be used in the Azure and other cloud vendors' self-hosted database benchmarks and can also be used to build on-prem database baselines for evaluating self-hosted or managed databases.
- Docker image for PostgreSQL benchmarking, targeting either a managed or self-hosted database, including auto-tuning of its parameters, such as the number of parallel threads and RPS used for querying, based on the target database/server specs; OLAP vs OLTP scenarios etc. (TBD after collab with benchANT). Note that this might also need a powerful instance.
- Automations to start managed and self-hosted databases along with their benchmarking clients, record results in the Navigator public data repositories.
- Monitor and keep track of the resource utilization of IaaS (TBD how to do that for DBaaS if we need to do so) to ensure the benchmarks are not limited by e.g. disk.
- Support SoftwareOne with the Performance Lab integrations.
- Vet results.
- Collaborate with other parties on the whitepaper.

benchAnt

Status: to be finalized, but likely to be a subcontractor of Spare Cores.

Team: Daniel Seybold + TBD

Tasks:

- License DBaaS Navigator methodology (seems to be already under the Apache license).
- Finalize the DBaaS qualitative data structure to be collected at the instance-type level (vs the current product level).
- Collect DBaaS qualitative data for a predefined list of vendors.

- Design and validate benchmarking methodology for PostgreSQL that can scale from small (e.g. 1 vCPU, 0.5 GiB RAM) to large (e.g. 500 vCPU, 8 TiB RAM) servers.
 - Implement/provide auto-tuning of database engine configs (see e.g. [aws/repro-collection](#) reference below).
 - Define/pick the datasets of various sizes and benchmarking scenarios (e.g. OLAP vs OLTP).
 - Define/pick queries used for benchmarking.
 - Implement/provide auto-tuning of benchmarking client parameters (e.g. RPS, warm-up tasks and timing).
 - Finalize benchmarking metrics to collect (e.g. throughput, latency).
 - Optionally, also discuss MySQL benchmarking as part of a future project.
- Vet results.
- Collaborate with other parties on the whitepaper.

Timeline

- June 16, 2026: Kickoff between SoftwareOne and Spare Cores
- June 18, 2026: Kickoff between Spare Cores and benchANT
- ...

Budget

- \$100,000: Spare Cores fixed fee for cloud infra engineering, orchestration, and Navigator updates.
- \$25,000-\$50,000: benchANT fixed fee for benchmarking methodology, and qualitative data.
- \$25,000-\$50,000: cloud compute and DBaaS costs.

Note that the cloud cost estimates can be heavily discounted with proper access to cloud credits, but otherwise, even just 1-1 hour of cloud instance runs costs:

Vendor	Servers	Min price	Avg price	Max price	Budget for 1 hr run
azure	1485	\$0.0063	\$6.06	\$206.26	\$8995
aws	1347	\$0.0048	\$6.38	\$389.0	\$8596
alicloud	992	\$0.0051	\$1.65	\$32.44	\$1635
gcp	529	\$0.0093	\$4.93	\$101.74	\$2605
vultr	173	\$0.003	\$0.8	\$5.76	\$138
upcloud	144	\$0.0045	\$1.51	\$36.0	\$217
ovh	72	\$0.0104	\$1.62	\$11.2	\$116

hcloud	25	\$0.0069	\$0.13	\$0.98	\$4
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NOT including the managed database instances, block storage (which will be a significant cost driver), and related cloud resources – expected overall 2x of the above estimates: up-to \$50k.

TBD how to structure the budget for the cloud costs, e.g. aiming for 100% credits VS fixed budget VS pass-through invoices.

Complexities

Already identified major technical hurdles requiring a significant amount of work to mitigate:

- Find “good-enough” block storage for each instance type to keep costs low without limiting compute performance with slow disks (potentially up to using RAID0 on IO2 volumes for larger instances to avoid an EBS bottleneck). Take into account the peak network and network storage bandwidth (published by vendors), and other known limiting factors (such as “Maximum supported I/O bandwidth” published by vendors) to save on benchmarking costs, otherwise use available max I/O performance (e.g. IO2 volumes).
- While trying to overprovision disk IO, it’s important to measure actual disk usage so that we can not only report on DB throughput and latency, but also on the max disk IO usage as a proxy what amount of provisioned IOPS capacity can be beneficial for a given instance type.
- Measure and record latency between DBaaS and IaaS products with different placement groups, so we can disclaim the related differences (e.g. when we can place two IaaS in the same rack, but the DBaaS is further away from the IaaS benchmarking client at a given vendor). E.g. extra benchmarks on measuring latency just running “SELECT 1” and documenting the process.
- Benchmarking scenarios, such as OLAP vs OLTP, small to large database sizes, concurrency, read VS write, batch COPY vs INSERT workloads etc – maybe use T-shirt sizes (S, M, L, XL workloads).

Limitations

1. Focus the performance measurements on **single-node** configurations: while DBaaS offers single-AZ, multi-AZ, and multi-region HA clusters, we will track these options only as features of the DBaaS products, and not measure failover time and degraded/improved performance in cluster mode. Similarly, not planning to benchmark cluster deployments of self-hosted databases as part of this project.
2. To keep costs at a reasonable level, focus performance measurements on a single **PostgreSQL version**, so benchmarking multiple major/minor versions of PostgreSQL is out of scope. Need to find a version that’s available at all supported DBaaS vendors.

3. Spare Cores is currently limited by **Azure quotas** and cannot start larger instances. Quotas needs to be increased, otherwise benchmarking will be limited to the smaller/mid-size server and DBaaS types.
4. Single-node **PostgreSQL does not scale well above 64 vCPUs**, so it might make sense to limit the list of IaaS products to ≤ 128 vCPU nodes.
5. Focusing on the **standard PostgreSQL engine** without custom extensions, such as PG-Storm for GPU support.

Outstanding Questions

- [Legal] Do we need consent from cloud providers to benchmark their DBaaS?
 - benchANT might have the related experience.
 - Hyperscalers' ToS seems to be fine if the methodology is transparent.
- [Legal] Do we need to let cloud providers know about the benchmarking efforts?
 - E.g. not to trigger alarms (as per AWS engineer).

References

- <https://www.postgresql.org/docs/current/pgbench.html>
- <https://github.com/akopytov/sysbench>
- <https://github.com/aws/repro-collection/tree/main/workloads>
- <https://github.com/aws/aperf>
- <https://github.com/benchANT/dbaas-navigator/tree/main/scoring>
- <https://heterodb.github.io/pg-strom/>